AMRUT NADGIR

amrutn@sas.upenn.edu
github.com/amrutn | Google Scholar

EDUCATION

Ph.D. Physics, University of Pennsylvania — Expected 2028

B.A. Physics, University of California, Berkeley — 2017-2021

PUBLICATIONS

Nadgir A, Balasubramanian V, Chaudhari P (2026) How Does Chain of Thought Decompose Complex Tasks? *International Conference on Machine Learning (ICML).* arXiv:2604.08872 [cs.LG]

Nadgir A, et al. Sudden, step-like learning curves in incremental Bayesian learners. *In preparation.*

Nadgir A, Thurston R, Larsen KA, Shivaram N, Brister M, Slaughter DS (2021) SILIA: software implementation of a multi-channel, multi-frequency lock-in amplifier for spectroscopy and imaging applications. *Meas. Sci. Technol.*

Jack A, Ferro LS, Trnka MJ, Wehri E, **Nadgir A**, Nguyenla X, et al. (2021) SARS-CoV-2 nucleocapsid protein forms condensates with viral genomic RNA. *PLoS Biol* 19(10): e3001425

EXPERIENCE

Ph.D. Student, Physics — Aug 2023 - Present
University of Pennsylvania
Advisors: Prof. Vijay Balasubramanian and Prof. Pratik Chaudhari — *Philadelphia, PA*

- Developing a scientific understanding of chain-of-thought reasoning in LLMs.
- Currently studying how local structure in the training data teaches an LLM to solve long-horizon tasks from very few training samples of bounded length.
  - Theoretical analysis is based on De Bruijn graphs.
  - Experiments require fine-tuning ∼8B parameter LLMs using low-rank adaptation (LoRA).
- Showed that training on reasoning traces that can be organized on a balanced prefix tree minimizes test error in LLMs (ICML 2026, arXiv:2604.08872).
  - Derived error scaling laws using Lipschitz (smoothness) arguments.
  - Trained small Transformers on synthetic logical deduction tasks and evaluated Qwen and DeepSeek models on mathematical reasoning benchmarks.
- Applied information geometry to show that incremental Bayesian learners exhibit sudden, step-like learning curves, matching dynamics observed in animals and deep neural networks.

Scientific Associate — Jul 2021 - Jul 2023
D.E. Shaw Research — *New York, NY*

- Trained a "triformer" neural network to sample molecular structures according to the Boltzmann distribution, accurately generating small-molecule geometries.
  - Training minimized a "free energy" loss: the entropy term was estimated by a kernel density estimator and the energy term was computed with a ForceField. The architecture was a Transformer applied to a triangular mesh.
- Built a tool creating an active learning loop for ForceFields, which explores ForceField dynamics using saddle-point search and adds the resulting samples back into the training data. This reduced how often ForceFields crashed in simulation.
- Improved the training process for in-house ForceFields.

Student Researcher Sept 2020 - May 2021
Neural Systems and Data Science Lab
Prof. Kristofer Bouchard *Berkeley, CA*

- Studied how training a recurrent neural network on toy problems reshapes its dynamics.
  - The dynamics became more nonlinear, measured by systematically varying the activation function and computing the predictive information in the resulting data.
  - The dynamics became non-normal (perturbations in preferred input directions cause large shifts in the output), measured using the Schur decomposition of the weight matrix.

Student Researcher April 2020 - Sept 2020
Yildiz Lab
Prof. Ahmet Yildiz *Berkeley, CA*

- Studied whether drug candidates could disrupt the liquid-liquid phase separation process that the COVID virus uses to repackage its RNA during replication (published).
  - Designed an image-analysis workflow to compute the size and number of phase-separated droplets formed at different concentrations of protein and drug candidate.

Student Researcher June 2018 - Dec 2020
Lawrence Berkeley National Lab
Dr. Daniel Slaughter *Berkeley, CA*

- Implemented a multi-channel lock-in amplifier in Python to extract the magnitude and phase of periodic features in multi-channel signal input (published, first author).

TECHNICAL SKILLS

Python, Bash, Slurm, PyTorch, HuggingFace Transformers, LoRA, NumPy/SciPy

AWARDS

National Science Foundation Graduate Research Fellowship Program (2024)

EXTRA-CURRICULAR ACTIVITIES

- Co-President, Penn Graduate Boxing Club (Aug 2024 - April 2026).
  - Managed dues, coaching, and training locations for a club of over 150 members.
  - Trained fighters for and helped organize Penn Fight Night, a charity boxing event drawing roughly 4,000 attendees and raising roughly $150,000 for the Boys and Girls Club of Philadelphia. Led our team to its first victory at the event in roughly 10 years.
- Executive Vice President, Pan-Asian Graduate Student Association, University of Pennsylvania (Aug 2024 - May 2025).
  - Started a lunch-and-learn program supporting student mental health and career readiness, and led a survey of the university's Asian community whose results were presented to the Vice Provost.